Mert Koca

Mapping neural circuits at nanometer scale, from bench to supercomputer.

LOCATION
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GITHUB
github.com/ikoshos-gland

> Highly self-driven and intellectually curious individual with a strong foundation in machine learning and neuroscience, working at the interface of expansion-microscopy connectomics and deep-learning segmentation. Adept at synthesizing complex information across disciplines into innovative, strategic solutions, and at managing long-term projects with a holistic approach that balances creative ideation with structured execution.

EXPERIENCE

Jul 2025 - Present
Istanbul, Turkey

Research Intern
KUTTAM, Koç University Research Center for Translational Medicine

FLAGSHIP RESEARCH

- Studied various **Expansion Microscopy (ExM)** protocols (ProExM, Magnify, and single-shot 20× ExM) to surpass the optical diffraction limit, enabling nanometer-scale imaging of neural tissue on confocal microscopes.
- Built and ran deep-learning **volume-segmentation pipelines** (Flood-Filling Networks, U-Net) for neural and vascular structures, working toward computer-based 3D brain mapping.
- Learned and operated **high-performance computing** on TRUBA (Turkey's national HPC center): SLURM batch scheduling, multi-GPU (JAX) training, and Apptainer/Singularity containers.

Oct 2024 - Feb 2026
Scholarship to Jul 2025

Embedded Machine Learning Scholar
TUSEB · Health Institutes of Turkey

- Interpreted **EMG signals** using LSTM+CNN, Random Forest, and Gaussian Process models.
- Developed a real-time signal-interpretation script.
- Produced data-analysis visualizations and animations in Python (Manim).

2024 - 2025
Quarter-finalist

Member
Easyegitim · İTÜ Çekirdek

- Combined **agentic tools** with specialized capabilities for students to achieve maximum efficiency from personalized education.

Jan 2024 - Jul 2024
National finalist

Team Member
Tez-Türk

- **EMG gesture classification** on the Ninapro DB5 dataset; built an MLP classifier for a prosthetic hand.

SELECTED PROJECTS

mertoshi.online
Personal website & agentic RAG platform, solo, end-to-end

- **Frontend:** React 19 + TypeScript + Vite + Tailwind + Spline 3D, deployed on Azure Static Web Apps.
- **Backend:** Python Azure Functions RAG using Azure OpenAI "On Your Data" + Azure AI Search (hybrid vector/semantic), secured with Managed Identity + Azure Key Vault.
- Features a custom AI assistant ("Lundo") and a document-intelligence + semantic-chunking indexing pipeline.

ESP32-S3 Bionic Hand *Real-time EMG-controlled bionic hand*

- Recognizes **11 hand gestures** in real time from **6 EMG sensors** (MyoWare) to drive a bionic prosthetic hand.
- On-device ML with **TensorFlow Lite Micro** on the ESP32-S3: Int8-quantized MLP, TD4 EMG features, and real-time IIR filtering in C/C++, under 300 ms end-to-end latency.

liconn-ffn-truba *Open-source HPC connectomics pipeline*

- Trains Google's **Flood-Filling Network** (JAX) on real expansion-microscopy connectomics data, scaling 1 → 4 GPUs on TRUBA.
- Public GitHub repository with an animated report.

EDUCATION

B.Sc., Molecular Biotechnology

Turkish-German University · Beykoz, Istanbul · *Graduated Jun 2026 · GPA 3.14 / 4.0*

Bachelor thesis (in German): light-microscopic connectomics beyond the diffraction limit: deep-learning volume segmentation comparing Flood-Filling Networks (FFN) vs. Local Shape Descriptors (LSD). Advisor: Prof. Dr. Atay Vural.

CERTIFICATION

Foundations of Quantum Mechanics

University of Colorado Boulder · *Coursera* · Verify

LANGUAGES

Turkish: Native · **German**: Advanced (C1, medium of instruction) · **English**: Fluent

COURSES

Quantum Biology

Onur Pusuluk · *Bilimler Köyü (Science Village) · 1 week*

Advanced Methods in Machine Learning

Çağatay Yıldız & Tolga Birdal · *Bilimler Köyü (Science Village) · 1 week*

PUBLICATIONS

- Coskuner-Weber O., Koca M., Uversky V. N. *"Molecular Crowding by Computational Approaches."* Springer Nature. DOI: 10.1007/978-3-032-03370-3_21
- Koca M. *"Kuantum-Klasik Bilinç Modelleri ve İnterdisipliner Perspektifler."* Quantum Dergisi, Issue 5 (qturkey.org).
- *Third manuscript in preparation.*

SKILLS

PROGRAMMING	**Python**, TensorFlow, JAX, NumPy, PyTorch (learning), Bash
MACHINE LEARNING	Deep learning (CNN, LSTM, MLP, Flood-Filling Networks, U-Net), Random Forest, Gaussian Processes, LangChain / LangGraph, RAG
HPC & CLOUD	**TRUBA HPC**, SLURM, multi-GPU training, Apptainer / Singularity, Azure (Functions, AI Search, OpenAI), Git / GitHub
IMAGING & BIO	Expansion microscopy, connectomics, image segmentation, pan-protein staining, immunofluorescence staining
TOOLS & VIZ	Manim, Matplotlib

INTERESTS

Photography (street & portrait) · running · interdisciplinary science · behavioral neuroscience.